

Sleep Project Final Report

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1. Introduction

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2. Methods

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Step by Step Explanation (add if wanted):

Set up *(add if wanted)*

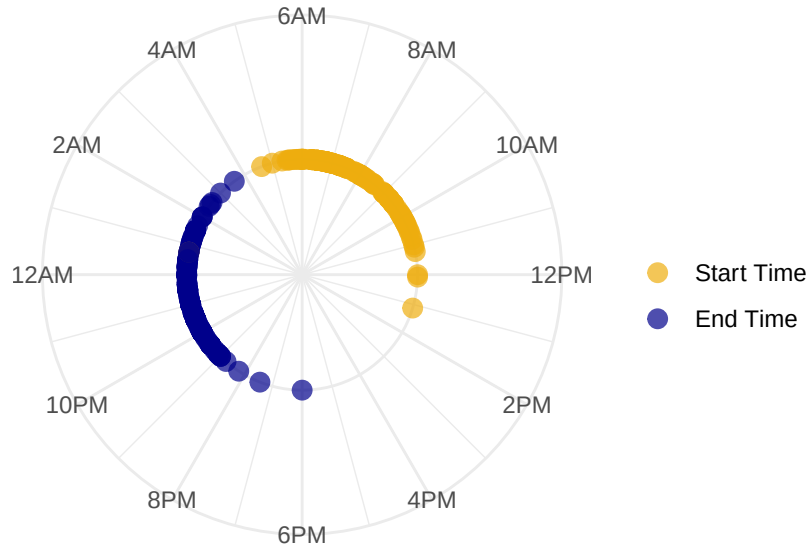
3. Results

Descriptive Results

Descriptive results: Through our upper quartile study on sleep quality we discovered that the top-quartile sleepers had a much longer sleep duration (*275 minutes higher*) as well as lower caffeine (*20% less*) use and less device time (*12 minutes less*) compared to sleepers who were not in the top-quartile. Unexpectedly, top sleepers also had higher anxiety ratings (*1.06 points higher*), more social media use(*18 minutes higher*), and less physical activity(*11.50 minutes less*).

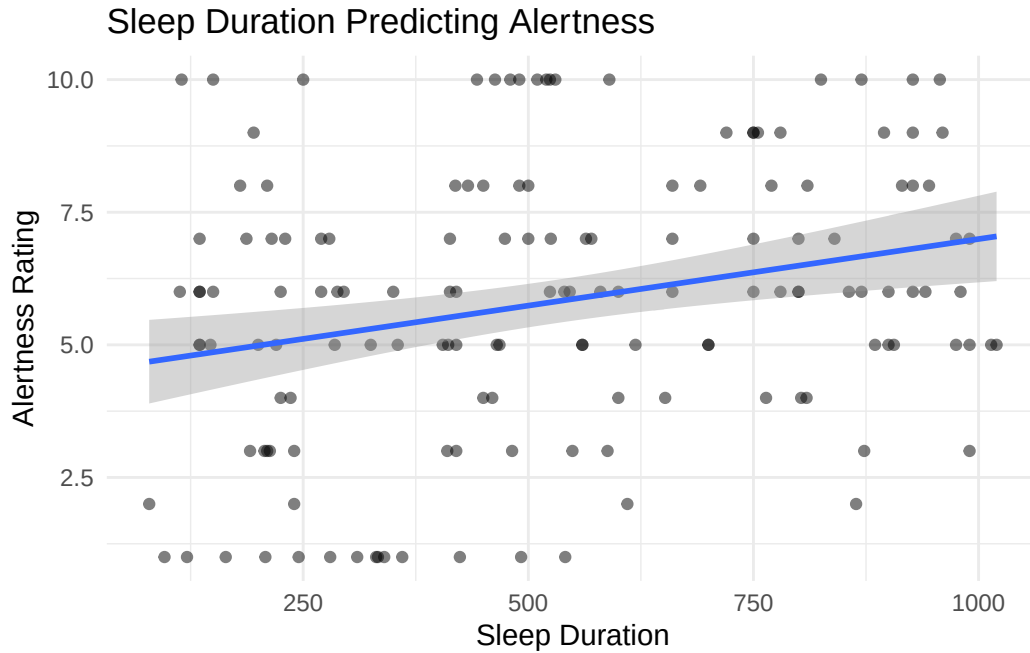
The average overall wake time was reported to be around 7:30 AM (7.665845) and the average sleep time was around 11:00 PM (23.26906).

Sleep Start and End Times (Clock View)

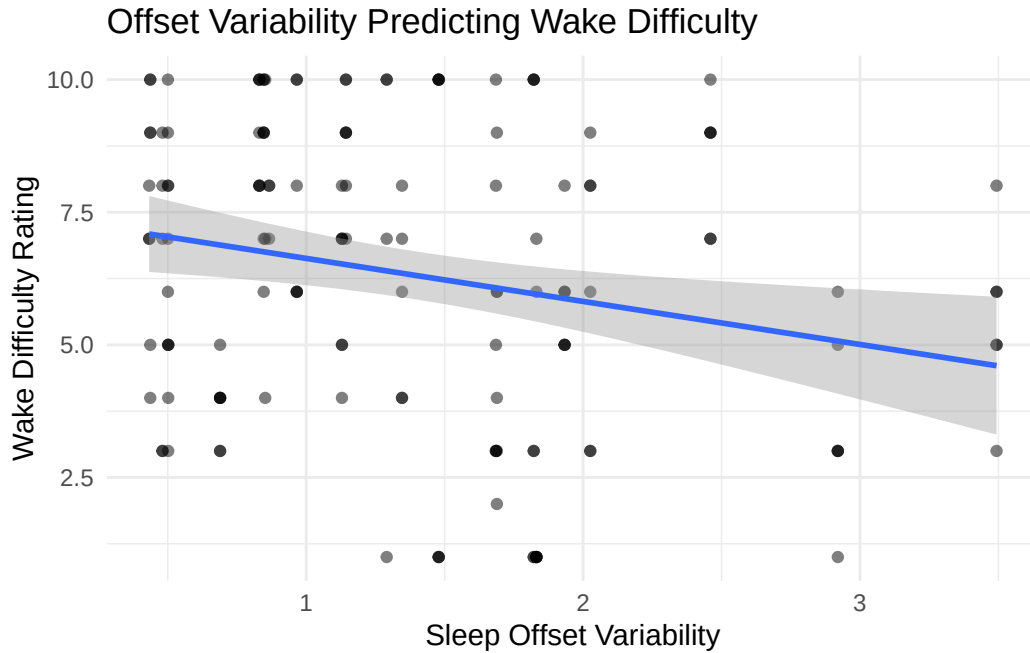


Regression Results

The regression study was based on the independent variables: onset time, offset time, onset variability, offset variability, sleep duration and the dependent variables: alertness, anxiety, wake difficulty, and sleep quality. Due to the recurring problem of heteroscedasticity violating the assumption of homoscedasticity in the linear regression analyses conducted for each dependent variable we used an HC3 heteroskedasticity-robust standard errors test to interpret coefficients more accurately. The results showed that sleep duration is a significant predictor for alertness ($p = 0.0096$, $= 0.0023$), indicating that longer sleep duration is associated with higher alertness. This made worked with our assumption that a longer/better sleep leads to a stronger sense of alertness throughout the day.



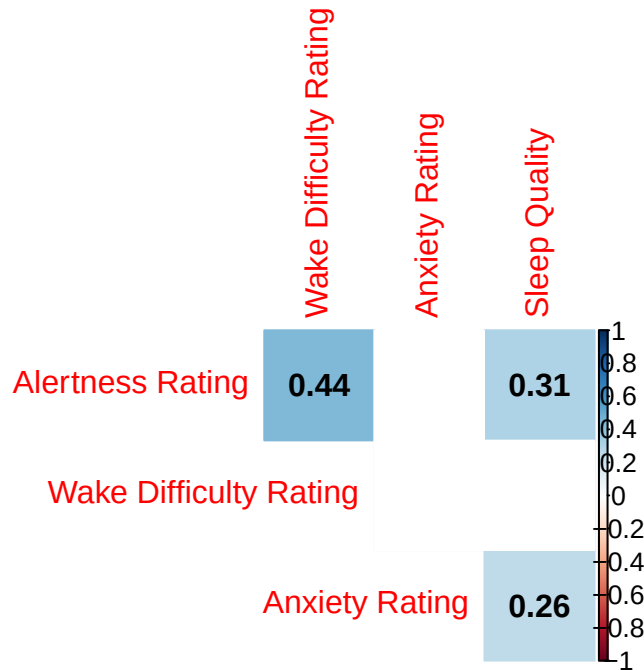
Another notable result is offset variability as a significant predictor for wake difficulty ($p=0.006$, $\beta = -0.720$), indicating that a greater inconsistency in wake time is associated with decreased difficulty. This was particularly interesting as it worked against our assumption that a more consistent wake time would lead to less difficult wake process aka lower wake difficulty ratings.



There were no other significant results for the remaining dependent variables: anxiety and sleep quality. However this is not to say that there are no predictors for these variables but just that there is further analysis that can be done to get a better grasp at what indicates anxiety and sleep quality reports from adolescents.

Correlation Results

We also looked into correlation between the dependent variables and found moderate correlations between alertness & wake difficulty ($r=0.42$), alertness & sleep quality ($r=0.31$), and anxiety & sleep quality ($r=0.30$).



All of the reported correlations are positive, showing us that the variables tend to increase together. The relationships are meaningful however they are not strong showing a range of around 0.3-0.4 for r . Higher alertness is associated with both easier wake times and better sleep quality which matches up with our assumptions on the effects of sleep quality on alertness. However, it also adds the additional perspective that an easier reported wake time is associated with higher alertness, something we didn't think to look into prior to this analysis. The correlation that anxiety has with sleep suggests that a higher anxiety is linked to a worse perceived sleep quality, this makes sense as someone who isn't as well rested may report a stronger sense of anxiety due to the lack of rest. These results indicate the dependent variables are related but not redundant meaning they capture overlapping but distinct aspects of the adolescent sleep experience within our data. Overall, the results support the idea that subjective sleep experiences/ratings are interconnected.

Mixed Effects Results

The data included 147 nights from 27 participants, within the data we noticed some trends although not statistically significance still peaked our interest. The trends we found were the association of more sleep than usual to a better reported sleep quality as well as a later sleep onset associated with a worse reported sleep quality. Though these are interesting trends it's important to emphasize that no predictors reached statistical significance in this study. The model assumptions were mostly satisfied showing no major variance issues and although there were some slight left-skew residuals the overall model fit is considered acceptable.

Takeaways

Sleep duration is the strongest consistent predictor within our study. However, most other sleep variables showed limited predictive powers. Some relationships exist between key outcomes, but casual predictors are weak overall.

4. Conclusions/Future Work

(paste conclusion here)

5. Appendix